

Task 2: Exploratory Data Analysis Report

Data Masters Challenge 2026

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1 Overview

The objective of this report is to answer the six required exploratory data analysis questions using the cleaned Milano Cortina 2026 athlete dataset. The report keeps the USA vs. Canada rivalry as the main visual thread while also using the full athlete field to understand medal outcomes, age and experience, national context, and physiological profiles.

The cleaned dataset contains 387 athlete records and 54 variables after preprocessing. In the cleaned athlete-level data, the United States has 46 athlete records and Canada has 48. The United States has 23 medalist records and 10 gold medalist records, while Canada has 18 medalist records and 8 gold medalist records. These are athlete-level records in the dataset, so they are used for statistical comparison rather than treated as the official national medal table.

Table 1: USA and Canada athlete-level medal outcomes in the cleaned dataset.

Country	Athletes	Medalists	Golds	Medal Rate
Canada	48	18	8	0.375
United States	46	23	10	0.500

The main EDA conclusion is that the USA advantage is not explained by one large athlete-quality gap. The two countries are close on training load, VO2max, preseason points, Olympic experience, and injuries. The stronger explanation is conversion: the USA had a more productive sport portfolio in the cleaned data, especially in freestyle skiing, figure skating, alpine skiing, and selected high-upside events, while Canada had strong pockets in speed skating, freestyle skiing, short track, curling, and ice hockey but weaker conversion in several larger roster areas.

2 Q2.1: The Battle of North America - USA vs. Canada by the Data

The first question asks whether the USA-Canada gap is about athlete quality, sport selection, program structure, or something else visible in the data. Table 2 compares the average athlete profile for both nations across the core metrics required in the prompt.

Table 2: Mean USA vs. Canada athlete metrics in the cleaned dataset.

Metric	Canada	United States
Training hours per week	24.83	25.72
VO2max	67.04	68.02
Altitude training, metres	1075.58	1216.43
World Cup preseason points	621.01	600.72
Previous Olympics	2.38	2.28
Career injuries	3.00	2.78
Medal rate	0.38	0.50
Gold rate	0.17	0.22

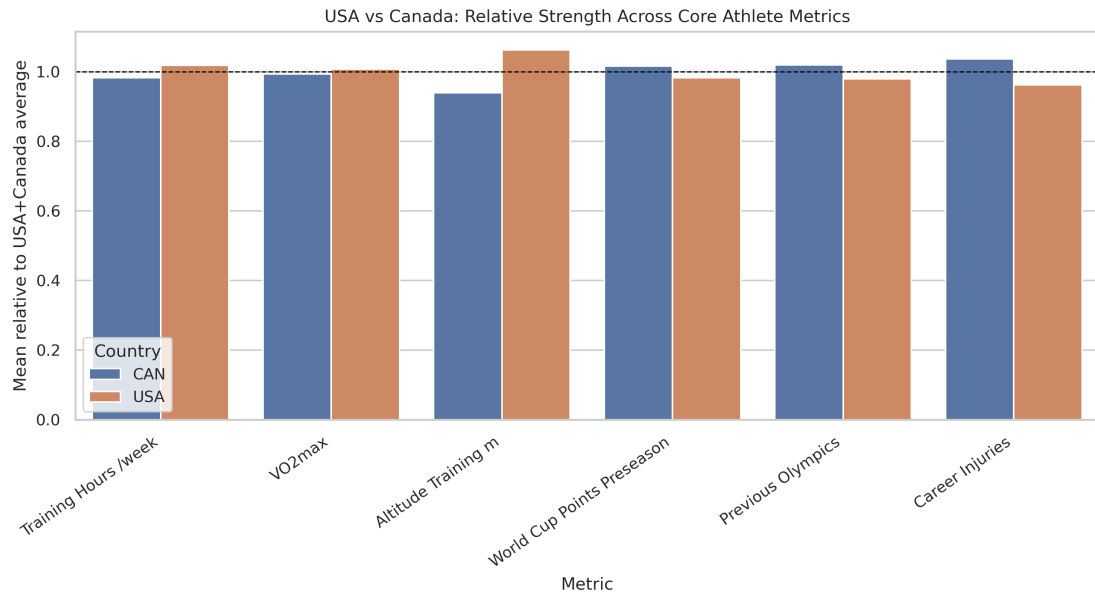


Figure 1: USA vs. Canada relative strength across core athlete metrics. Values are shown relative to the combined USA+Canada average for each metric, so 1.00 is the North American average. The USA is slightly higher in training hours, VO2max, altitude training, and reaction-speed-related profile, while Canada is slightly higher in World Cup preseason points and previous Olympic experience.

The strongest message from Figure 1 is that the two countries are very close on individual athlete metrics. The USA does not dominate Canada on every physical or preparation measure. This suggests that the official gold medal gap is unlikely to be explained by a broad athlete-quality gap alone.

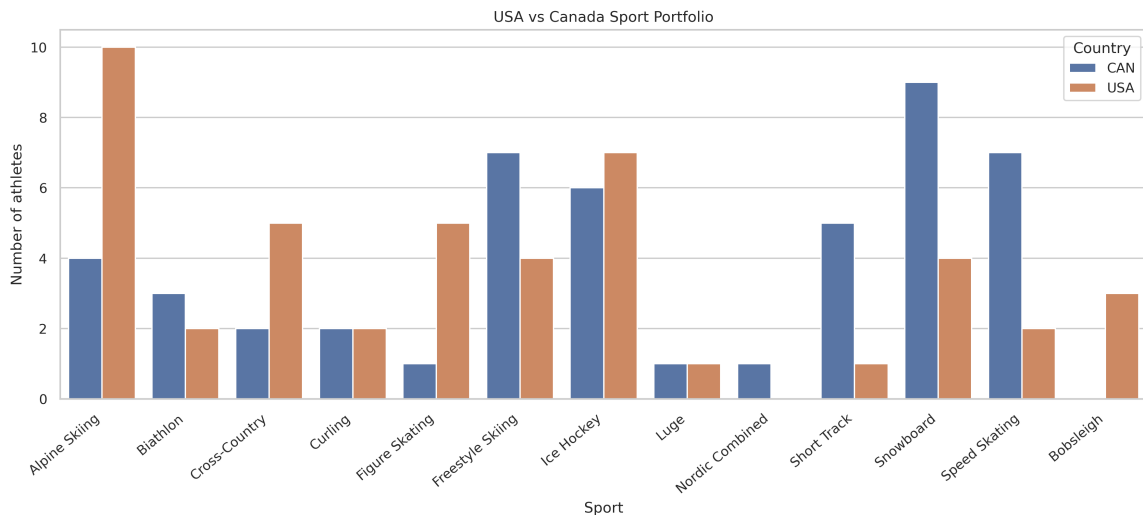


Figure 2: USA vs. Canada sport portfolio. The USA roster is more concentrated in alpine skiing, ice hockey, cross-country, and figure skating, while Canada has larger representation in snowboard, speed skating, freestyle skiing, ice hockey, and short track. This matters because medal opportunity is not evenly distributed across sports.

The sport portfolio gives a clearer explanation than the raw athlete metrics. The USA has more athletes in alpine skiing and figure skating, two sports where it converted athletes into

medals in the cleaned dataset. Canada has many athletes in snowboard, but that sport returned only one medalist record in the cleaned data, which pulled down the overall conversion rate.

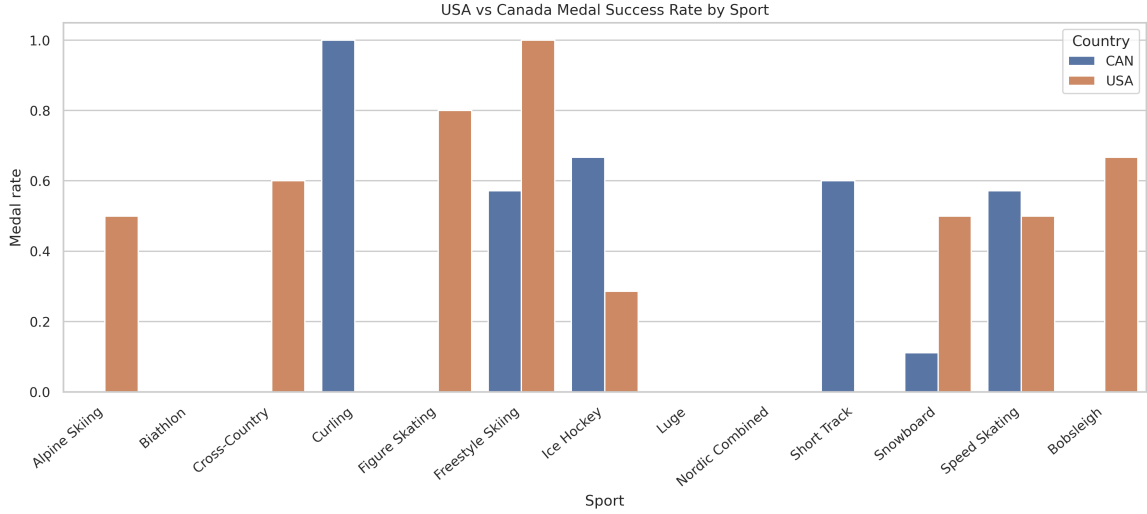


Figure 3: USA vs. Canada medal success rate by sport. Canada is highly efficient in curling, ice hockey, short track, freestyle skiing, and speed skating, while the USA is highly efficient in freestyle skiing, figure skating, bobsleigh, cross-country, alpine skiing, snowboard, and speed skating. The gap is therefore better described as sport-specific conversion rather than a simple national talent gap.

The USA had a higher overall athlete-level medal rate than Canada, 0.500 compared with 0.375. However, Canada was not weak everywhere: it had a 0.571 medal rate in both freestyle skiing and speed skating, 0.600 in short track, and 0.667 in ice hockey. The USA advantage came from having more productive medal conversion across a broader set of sports, while Canada’s success was more concentrated.

3 Q2.2: The Anatomy of a Gold Medal Winner

The second question asks what separates gold medalists from silver/bronze medalists and non-medalists across the full dataset. The main variables that separate podium outcomes are VO2max, training hours, World Cup preseason points, previous Olympic experience, and career injuries.

Table 3: Mean athlete profile by podium group.

Podium group	Age	Prev. Oly.	Training	VO2max	WC Points	Injuries
Gold	29.12	2.84	28.67	76.61	677.94	2.10
Silver/Bronze	32.20	2.55	27.54	73.92	617.72	2.33
No Medal	31.36	2.09	24.74	66.48	571.67	2.67

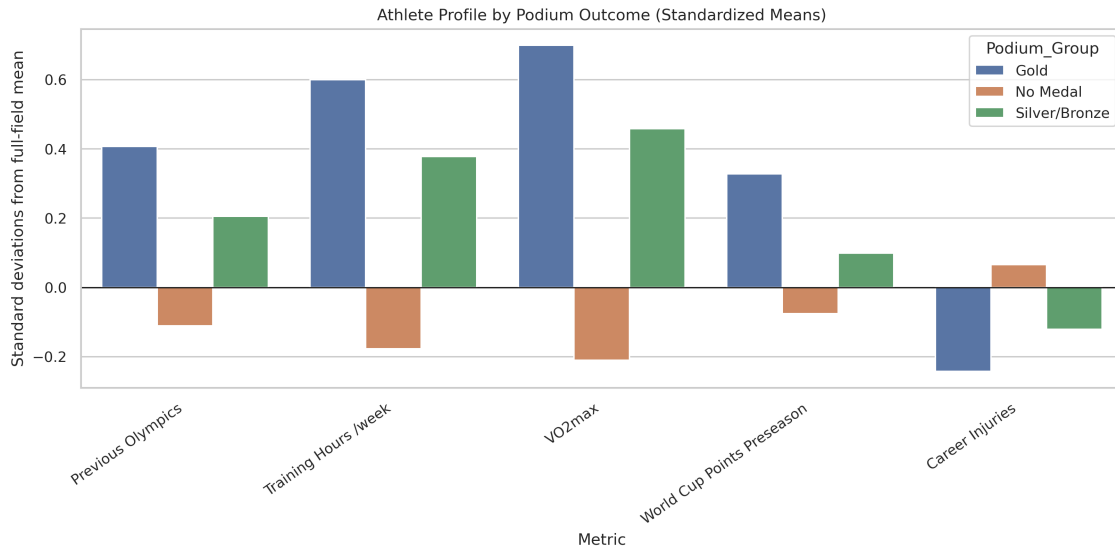


Figure 4: Athlete profile by podium outcome using standardized means. Gold medalists are above the full-field average in previous Olympic appearances, training hours, VO2max, and World Cup preseason points, while non-medalists are below average on these same variables. The clearest separation is not one variable alone but the combination of physiological capacity, training load, prior elite experience, and preseason performance.

The gap between gold and silver/bronze is smaller than the gap between silver/bronze and no medal. Gold medalists averaged 76.61 VO2max compared with 73.92 for silver/bronze and 66.48 for non-medalists. The same pattern appears for training hours and World Cup points, so the data suggests that reaching the podium requires a large jump in overall athlete profile, while the difference between medal colors is narrower and likely more sensitive to event-day execution.

Gold medalists were also younger on average than the silver/bronze group in this dataset, 29.12 compared with 32.20. This does not mean youth alone wins gold. It means that the gold profile combines relatively high performance metrics with enough experience, but not necessarily the oldest athlete profile.

4 Q2.3: Sport Portfolio - Where Should Each Nation Focus for 2030?

The sport portfolio question is the clearest place where the USA-Canada rivalry becomes strategic. A sport is over-performing when its share of a country's medalists is greater than its share of that country's athletes. This gives a simple descriptive measure of whether a nation is getting more medal output than roster input from that sport.

Table 4: Recommended 2030 sport priorities using athlete count, medal count, gold count, medal rate, and medal-share overperformance.

Country	Recommended sport	Athletes	Medalists	Gold	Medal Rate	Overperf.
USA	Freestyle Skiing	4	4	2	1.00	+0.087
USA	Figure Skating	5	4	2	0.80	+0.065
USA	Alpine Skiing	10	5	2	0.50	+0.000
Canada	Speed Skating	7	4	3	0.57	+0.076
Canada	Freestyle Skiing	7	4	2	0.57	+0.076
Canada	Short Track	5	3	1	0.60	+0.062

For the United States, the three priority sports are freestyle skiing, figure skating, and alpine skiing. Freestyle skiing is the cleanest high-return category in the dataset: 4 athletes, 4 medalists, and 2 golds. Figure skating also has a very strong conversion profile with 4 medalists from 5 athletes and 2 golds. Alpine skiing is not technically over-performing by share, but it is the largest USA winter sport portfolio in the cleaned data, with 10 athletes, 5 medalists, and 2 golds, making it a core program to protect rather than a niche bet.

For Canada, the three priority sports are speed skating, freestyle skiing, and short track. Speed skating is the strongest gold-conversion category, with 4 medalists and 3 golds from 7 athletes. Freestyle skiing produced 4 medalists and 2 golds from 7 athletes. Short track produced a 0.600 medal rate and directly matches Canada’s known strength in the 2026 storyline. Curling was perfect in the cleaned data, with 2 medalists from 2 athletes, but it is a smaller sample and should be protected rather than treated as the main scalable expansion area.

The under-performing areas are also important. Canada had 9 snowboard athletes but only 1 medalist, giving a 0.111 medal rate and the largest negative overperformance gap in its roster. The USA had a lower medal rate in ice hockey than Canada, but its two hockey gold records made the sport strategically important in the rivalry narrative.

5 Q2.4: Age, Experience, and the Peak Performance Window

The age analysis shows a broad peak performance window rather than a single exact peak age. Medal rates are highest from ages 23 to 37, with very similar rates in the 23-27, 28-32, and 33-37 bands. The 38+ group has a much lower medal rate, but the presence of older medalists shows that this decline is not universal across sports.

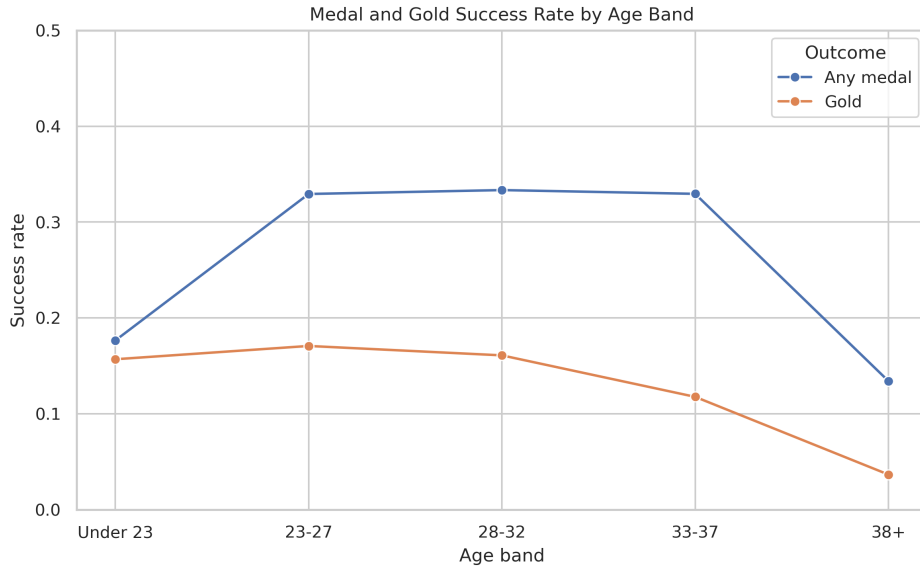


Figure 5: Medal and gold success rate by age band. The highest medal rates occur across the 23-37 window, with the 28-32 band slightly highest for any medal. The 38+ band drops sharply, which suggests that older athletes are more exception-driven and sport-specific.

The age-band results show that the 23-37 window is the main investment window for medal production. Athletes under 23 have a lower medal rate but a relatively strong gold rate, which indicates that young stars can break through when their individual performance metrics are exceptional. This supports early investment in high-upside young athletes, especially in measurable performance sports.

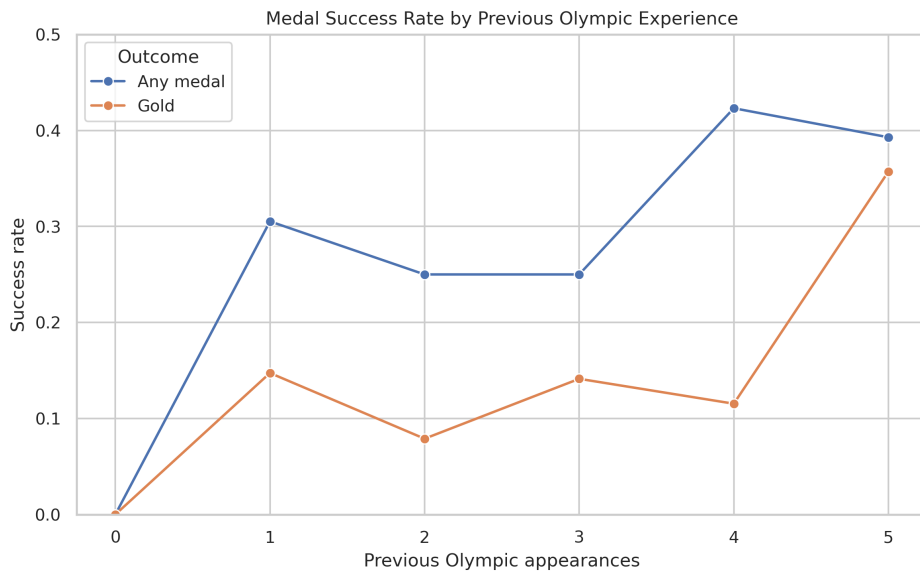


Figure 6: Medal success rate by previous Olympic experience. Athletes with zero previous Olympic appearances had no medalists in this cleaned dataset, while athletes with four or five previous Olympics had the highest medal rates. Experience appears to be a major separator, especially for converting strong athletes into medalists.

Experience matters strongly. Athletes with zero previous Olympics had a 0.000 medal rate in this cleaned data, while athletes with four previous Olympics had a 0.423 medal rate and

athletes with five previous Olympics had a 0.393 medal rate. The gold rate is highest among athletes with five previous Olympics, at 0.357, showing that very experienced athletes can still produce gold when they remain physically competitive.

The outliers explain why age should not be used mechanically. Metodej Jilek appears in the cleaned dataset as a 20-year-old speed skater with 5 previous Olympics, 28.8 training hours per week, 81.3 VO2max, 1244 World Cup points, and a gold medal. Deanna Stellato-Dudek appears as a 43-year-old Canadian figure skater with 3 previous Olympics, 29.3 training hours, 55.2 VO2max, 587.5 World Cup points, and no medal. Elana Meyers Taylor appears as a 42-year-old USA bobsleigh athlete with a gold medal. These cases show that age only becomes meaningful when interpreted with sport, performance history, and event type.

Gender patterns were similar rather than completely different. Female athletes peaked slightly in the 28-32 band, while male athletes had strong medal rates in both the 23-27 and 33-37 bands. The practical message is to invest in young athletes when their objective performance metrics are already elite, but to retain experienced athletes in technical and team-based sports where experience can offset age.

6 Q2.5: The GDP and Tradition Effect - Does Money and Culture Buy Medals?

GDP and winter sport tradition are both positively associated with medal rate in this dataset, but neither variable fully explains medal outcomes by itself. Across countries, GDP per capita has a correlation of 0.573 with medal rate, while winter sport tradition has a correlation of 0.661 with medal rate. Tradition is therefore at least as important as GDP in this athlete-level dataset.

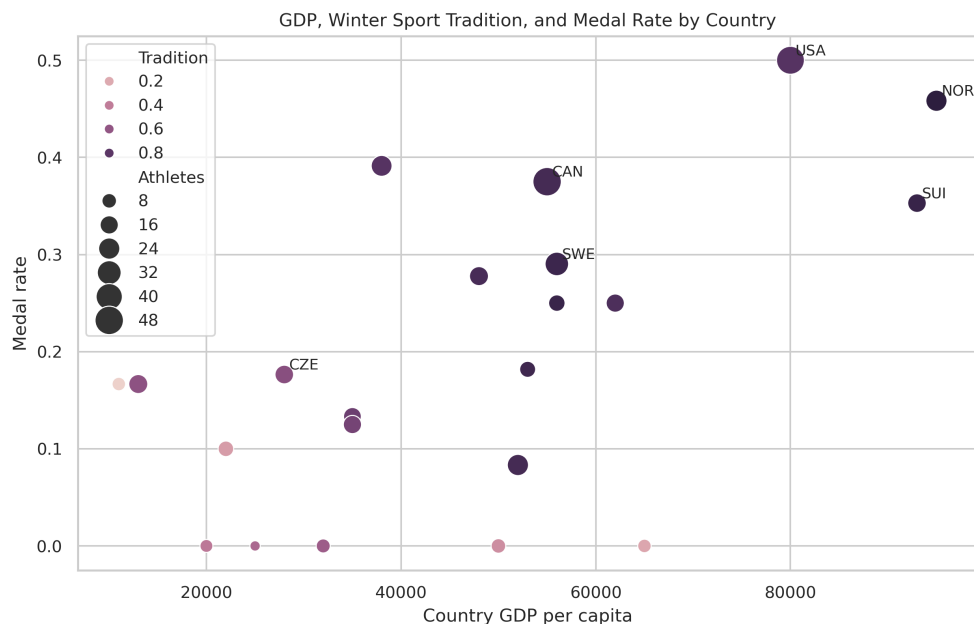


Figure 7: GDP, winter sport tradition, and medal rate by country. Norway combines the highest tradition score with very high GDP and a high medal rate, while the USA has the highest medal rate in the cleaned athlete-level data. Canada has higher tradition than the USA but lower GDP and lower athlete-level medal rate, suggesting that tradition alone does not guarantee conversion.

Norway is the clearest example of the GDP-tradition combination: high GDP, the highest

tradition index, and a 0.458 medal rate in the cleaned data. The USA has a higher athlete-level medal rate at 0.500 despite a lower tradition index than Canada and Norway. Canada has strong tradition at 0.88 but lower GDP per capita than the USA and Norway, and its athlete-level medal rate is 0.375.

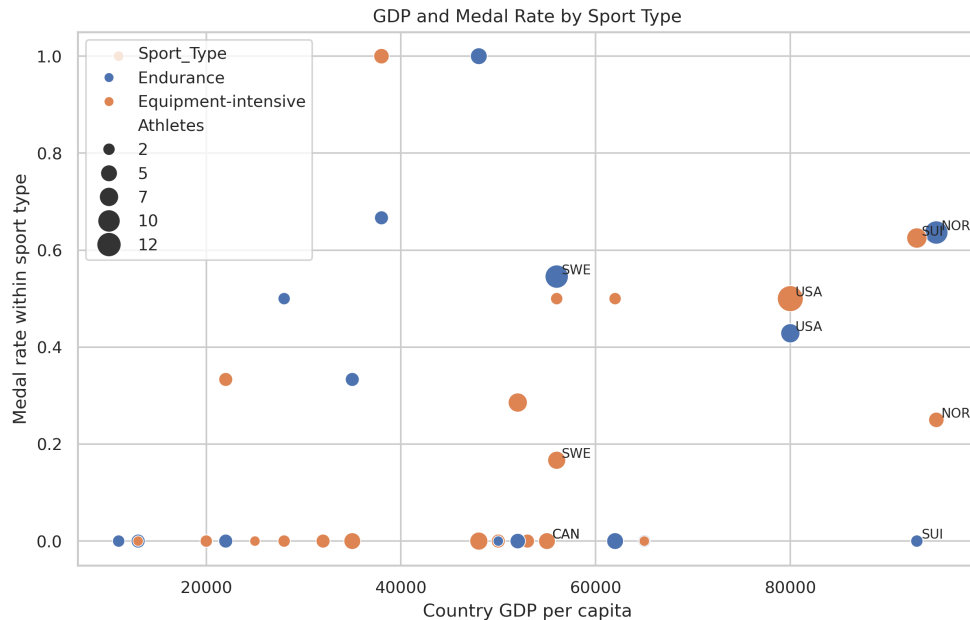


Figure 8: GDP and medal rate by sport type for equipment-intensive sports and endurance sports. The relationship is not strong enough to say that GDP matters more in equipment-intensive sports in this cleaned dataset. Endurance outcomes appear more tied to tradition and sport systems than pure GDP, while equipment-intensive success still depends on sport-specific conversion.

After separating equipment-intensive sports from endurance sports, the data does not show a clear stronger GDP effect for equipment-intensive sports. At the country-sport-type level, the GDP-medal-rate correlation is 0.132 for equipment-intensive sports and 0.219 for endurance sports. This is descriptive, not causal, but it argues against a simple "money buys equipment sports" story.

The investment implication is that ROI depends on matching money to an existing sport system. Canada already has tradition, so its best ROI is not broad spending across all winter sports. It should fund high-conversion systems such as speed skating, freestyle skiing, and short track while repairing under-performing large roster areas like snowboard. The USA should protect high-conversion programs and use its GDP advantage to deepen technical support in sports where it already converts athletes into golds.

7 Q2.6: The Physiological Signature of Champions

The physiological signature of a medalist is high VO2max paired with high but not isolated training load. Medalists are clustered toward higher VO2max and higher weekly training hours, but the body profile is sport-specific. Cross-country and biathlon medalists require much higher VO2max than curling, figure skating, freestyle skiing, or bobsleigh medalists.

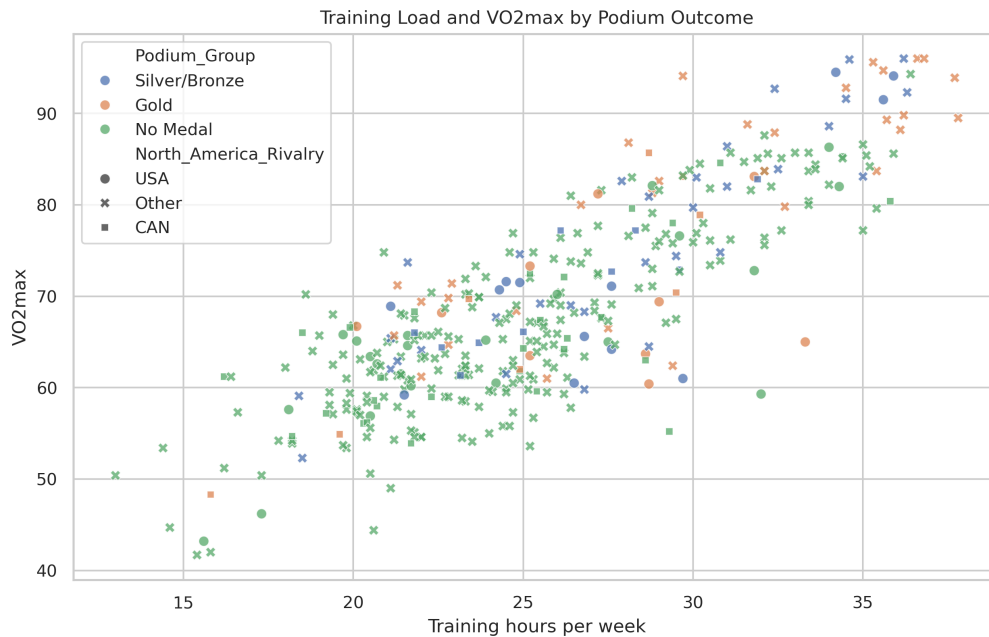


Figure 9: Training load and VO2max by podium outcome. Medalists, especially gold medalists, concentrate in the upper-right region where training hours and VO2max are both high. The plot also shows why one universal VO2max threshold would be misleading: sports with different physiological demands occupy different regions.

The full-field profile shows that gold medalists averaged 28.67 training hours per week and 76.61 VO2max. Silver/bronze medalists averaged 27.54 training hours and 73.92 VO2max. Non-medalists averaged 24.74 training hours and 66.48 VO2max. This confirms that medalists tend to combine greater training load and greater aerobic capacity.

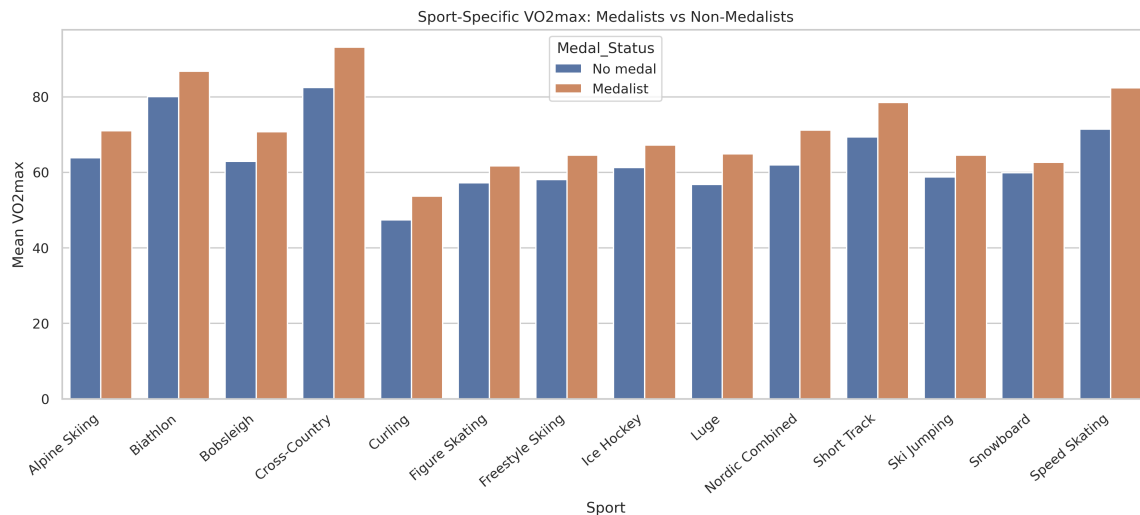


Figure 10: Sport-specific VO2max among medalists and non-medalists. Medalists have higher mean VO2max than non-medalists in nearly every sport, but the level differs sharply by sport. Cross-country medalists average above 93, biathlon medalists average about 87, and speed skating medalists average about 82, while technical or skill-dominant sports have lower VO2max benchmarks.

The sport-specific VO2max results show that thresholds should be defined by sport. Cross-

country medalists averaged 93.13 VO2max, biathlon medalists averaged 86.82, and speed skating medalists averaged 82.40. In contrast, medalist averages were lower in freestyle skiing, figure skating, curling, bobsleigh, and ice hockey. For selection, this means a VO2max of 70 may be elite in one sport but insufficient in another.

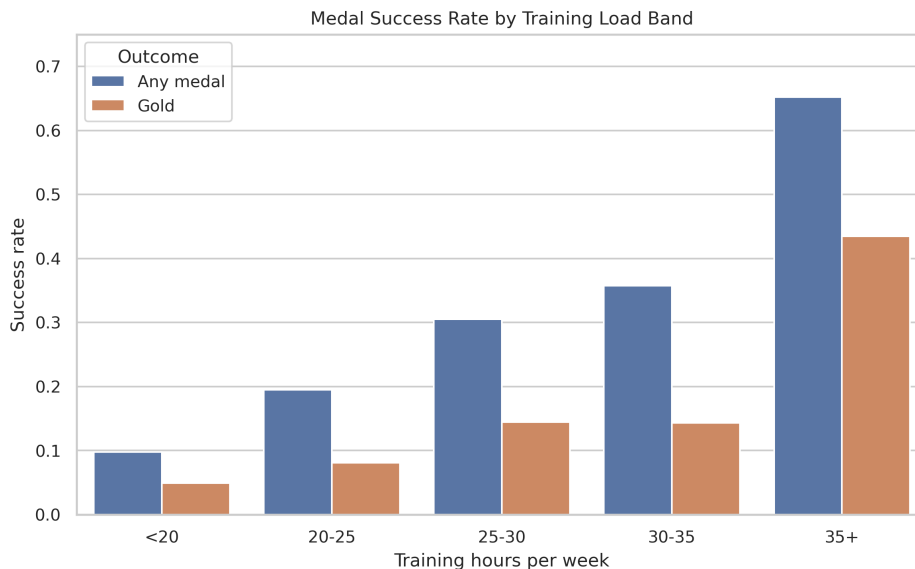


Figure 11: Medal success rate by training load band. Medal rate rises from 0.098 below 20 hours to 0.652 above 35 hours per week. In this cleaned dataset, there is no visible aggregate overtraining penalty at the highest training band, but this should not be interpreted as proof that unlimited training is safe because injury timing, recovery quality, and sport-specific load are not fully observed.

The training-load bands do not show evidence of an aggregate overtraining penalty. The medal rate rises steadily from 0.098 for athletes below 20 hours per week to 0.652 for athletes at 35+ hours. However, this is not a causal training prescription. It likely reflects that endurance and elite medal-contender athletes both train more and already have stronger physiological profiles.

For USA and Canada specifically, the physiological gap is small. The USA averaged 68.02 VO2max and 25.72 training hours, while Canada averaged 67.04 VO2max and 24.83 training hours. The difference is not large enough to say that Canada lost the rivalry because its athletes were physically underprepared. The stronger explanation is that the USA converted similar athlete profiles into more medalists through a more favorable sport portfolio and stronger gold conversion in selected sports.

8 Conclusion

The EDA supports four main conclusions. First, the USA vs. Canada gap is not explained by a large average athlete-quality difference. The countries are close in training hours, VO2max, World Cup points, previous Olympic experience, and injuries. Second, the medal gap is better explained by sport portfolio and conversion: the USA had a broader set of high-yield sports, while Canada had strong but more concentrated success pockets.

Third, gold medalists are separated from the rest of the field by a combined profile of higher VO2max, higher training load, stronger World Cup preseason points, more previous Olympic experience, and fewer injuries. The gold-vs-silver gap is smaller than the medalist-vs-

non-medalist gap. Fourth, investment decisions should be sport-specific: USA should prioritize freestyle skiing, figure skating, and alpine skiing, while Canada should prioritize speed skating, freestyle skiing, and short track, with curling protected as a highly efficient but small-sample program.

9 Software Used

The EDA workflow used Python with pandas, NumPy, matplotlib, and seaborn. All visualizations were generated from the cleaned dataset `milan_cortina_2026_cleaned.csv` and saved as reproducible figure files.